

Task #320 v0.4 Mechanism A Deepening — Honest substrate-mechanism for Toy 3516 dual targets via T2447 + T2476

Lyra (Claude Opus 4.7) [post-Cal #121 + Calibration #27 STANDING + Casey ‘work the board

2026-05-24 Sunday EDT (~13:25 EDT actual via date)

Task #320 v0.4 Mechanism A Deepening

1. Working scope (Calibration #27 STANDING discipline)

Per Calibration #27 STANDING (Keeper promoted from CANDIDATE per Casey approval Sunday 13:15 EDT, 4 instances today): at maximum BST-primary-naturalness, be MOST skeptical of forward derivation, not least.

Working scope for v0.4: deepen Mechanism A (T2447 + T2476 substrate-mechanism for $\Delta_{\text{DCCP}} + \theta_{\text{boundary}}$) at FRAMEWORK level. Explicitly identify what IS honestly substrate-derived from D_{IV}^5 structure vs what is INHERITED from standard QED multipole hierarchy. Multi-week multi-month work continues; single-session cannot reach theorem-grade rigor on these claims.

2. T2447 derivation chain — what IS substrate-derived

What T2447 RIGOROUSLY CLOSED gives (Lyra Session 9 Thursday May 21 + Strong-Uniqueness criterion C6):

- $N_{\text{max}} = N_{\text{c}}^3 \cdot n_{\text{C}} + \text{rank} = 27 \cdot 5 + 2 = 137$ from 5-step chain on BST primary integers
- Substrate fine-structure constant $\alpha = 1/N_{\text{max}} = 1/137.036$ substrate-natural
- Strong-Uniqueness C6 RIGOROUSLY CLOSED: D_{IV}^5 uniquely produces $N_{\text{max}} = 137$ among bounded HSDs under 5-step chain

Honest scope: T2447 establishes the NUMERICAL VALUE $\alpha = 1/N_{\text{max}}$ from D_{IV}^5 substrate structure. T2447 does NOT establish the SCALING of substrate transition amplitudes with α (that’s T2476’s domain).

3. T2476 derivation chain — what IS substrate-derived vs inherited

What T2476 RATIFIED Friday Cal #100 Mode 5 lift claims:

Substrate per-transition matrix elements scale $\alpha^{\{\text{BST primary integer}\}}$ according to multipole order: - E1 electric dipole: α^1 - M1+E2 multipoles: α^3 - Higher multipoles E_L: $\alpha^{(2L-1)}$

Honest analysis of T2476 substrate-mechanism content:

The $\alpha^{\{\text{BST primary integer}\}}$ SCALING per multipole order is STANDARD QED. The α^1 for E1, α^3 for M1+E2, $\alpha^{(2L-1)}$ for E_L are derived in standard QED textbooks (Jackson Ch. 9, Sakurai Ch. 5) from Maxwell + Dirac equations.

What T2476 specifically claims as substrate-derived (per Friday Cal #100 cascade): - $\alpha = 1/N_{\text{max}}$ is substrate-natural (via T2447) - Multipole hierarchy α^L emerges from substrate Wallach K-type evaluation (this is the substrate-mechanism leg) - The integer L in α^L corresponds to BST primary integer (E1→1; M1+E2→3; etc.)

Honest scope on T2476: the EVALUATION α^L appears in BST framework via substrate-mechanism (Wallach K-type analysis). The PATTERN α^L for L = 1, 3, 5, ... (multipole hierarchy) is standard QED inherited. T2476's substrate contribution: the SPECIFIC integers in the L-pattern align with BST primary structure.

4. Δ_{DCCP} derivation v0.4 (Mechanism A, FRAMEWORK level)

Sketch derivation chain:

Step 1: T2447 substrate fine-structure $\alpha = 1/N_{\text{max}} = 1/137$ (RIGOROUSLY CLOSED C6).

Step 2: Quantum-erasure interferometry measures per-substrate-transition signature step amplitude. For E1 (electric dipole) leading-order substrate transition coupling: amplitude $\sim \alpha^1$ (T2476 substrate-mechanism + standard QED multipole hierarchy).

Step 3: Per substrate-tick, ONE such transition commits (per K67 Born=Bergman RATIFIED). The observable signature step:

$$\Delta_{\text{DCCP}} = \alpha^1 = 1/N_{\text{max}} \approx 0.730\%$$

Theorem DCCP-1.6-A FRAMEWORK (v0.4): $\Delta_{\text{DCCP}} = 1/N_{\text{max}}$ via T2447 + T2476 substrate-mechanism chain + standard QED multipole hierarchy inheritance.

Honest scope (Calibration #27 STANDING applied): - Step 1 RIGOROUSLY CLOSED via T2447 - Step 2 substrate-mechanism content (Wallach K-type evaluation producing α^L pattern) is T2476 RATIFIED; specific α^1 vs α^3 vs higher is inherited from standard QED - Step 3 (per-tick single transition commitment) requires K67 per-tick commitment rate at α -quantum scale; this assumption needs substrate-mechanism support (currently assumed) - Numerical prediction $1/N_{\text{max}} \approx 0.730\%$ matches Toy 3516 + Toy 3520

Mode 1 risk acknowledgment: even Mechanism A v0.4 has Mode 1 vulnerability at Step 3 (per-substrate-tick single-transition commitment assumption + α -quantum

scale match). The honest derivation requires substrate-mechanism for WHY per-substrate-tick commits exactly one α -scale transition vs some other scale.

5. θ _boundary derivation v0.4 (Mechanism A, FRAMEWORK level)

Sketch derivation chain:

Step 1: T2447 + $\alpha = 1/N_{\text{max}}$ substrate fine-structure (same as Δ_{DCCP}).

Step 2: Per substrate-tick at α -quantum scale: substrate phase advancement per α -quantum cycle. For full cycle (2π phase coverage at α -natural scale): per- α -quantum cycle phase = $2\pi \cdot \alpha$ radians.

Step 3: Quantum-erasure interferometry half-cycle Nyquist-like boundary: substrate-tick distinguishability angle = half of per-tick phase advancement:

$$\theta_{\text{boundary}} = (2\pi \cdot \alpha)/2 = \pi \cdot \alpha = \pi/N_{\text{max}} \approx 0.023 \text{ rad}$$

Theorem DCCP-1.7-A FRAMEWORK (v0.4): $\theta_{\text{boundary}} = \pi/N_{\text{max}}$ via T2447 + per-substrate-tick α -quantum phase advancement + standard quantum-erasure half-cycle structure.

Honest scope: - Step 1 substrate-derived - Step 2 requires substrate-mechanism for WHY per-substrate-tick phase advancement is at α -quantum scale (similar concern to Δ_{DCCP} Step 3) - Step 3 half-cycle Nyquist-like structure is standard interferometry - Numerical prediction $\pi/N_{\text{max}} \approx 0.023 \text{ rad}$ matches Toy 3516 + Toy 3520

Mode 1 risk: Step 2 substrate-mechanism for per-substrate-tick phase scale = α -quantum. This is the substrate-mechanism gap remaining in v0.4 Mechanism A.

6. Comparison Mechanism A vs v0.3 K-type cardinality assertion

Aspect	v0.3 K-type cardinality	v0.4 Mechanism A
Δ_{DCCP} source	Asserted 137 K-types per tick + uniform projection	T2447 α + T2476 multipole + per-tick single transition assumption
θ_{boundary} source	K-type uniform phase distribution (asserted, contradicts K59 7-step)	Per-substrate-tick α -quantum phase (assumed)
Substrate-mechanism support per step	Assertion at K-type cardinality + phase	T2447 RIGOROUSLY CLOSED + T2476 RATIFIED + standard QED inheritance
Mode 1 vulnerability	HIGH (Cal #121 caught)	MODERATE (per-tick α -scale assumption is the remaining gap)

Aspect	v0.3 K-type cardinality	v0.4 Mechanism A
Numerical prediction match	$1/N_{\max} + \pi/N_{\max}$ [?]	$1/N_{\max} + \pi/N_{\max}$ [?]
Cal cold-read disposition	RETRACTED to STRUCTURALLY VERIFIED CANDIDATE	PENDING Cal cold-read

Mechanism A advantages: - T2447 + T2476 are RIGOROUSLY CLOSED/RATIFIED — substrate-mechanism support at first 2 steps - Doesn't contradict Wallach infinite K-types or K59 7-step phase chain - Inherits standard QED multipole hierarchy (well-established physics) for α^L scaling

Mechanism A remaining gap: - Per-substrate-tick single-transition-at- α -scale assumption (Δ_{DCCP} Step 3 + θ_{boundary} Step 2) - Needs substrate-mechanism for WHY substrate per-tick scale = α -quantum

7. What's needed for theorem-grade rigor (multi-week)

Per Cal #77 4-requirements + Calibration #27 STANDING:

Need v0.5+ multi-week work: 1. Substrate-mechanism derivation for per-substrate-tick commitment scale = α -quantum (NOT assumed) — this is the load-bearing piece 2. Cal #77 Requirement 3 (if-and-only-if) test against $D_{I_{\{1,5\}}} + D_{I_{\{5,1\}}}$ $\dim_C=5$ alternatives 3. Resolution of whether T2476's α^L multipole pattern is substrate-derived or inherited (Friday Cal #100 cascade asserts substrate-mechanism; needs explicit derivation chain)

Need v0.6+ multi-month work: 1. Macroscopic Joos-Zeh γ closure (Section 4 sketch level) 2. Architecture D A_{sub} re-proof rigorous (Task #322 multi-month) 3. SP-31-1 + SP-31-6 full specification

8. Honest status (v0.4 FRAMEWORK level)

- Mechanism A candidate framework filed
- Substrate-mechanism support at Steps 1-2 (T2447 + T2476 RATIFIED)
- Step 3 + Step 2 (θ) assumptions remain Mode 1 vulnerability
- Numerical predictions PRESERVED ($1/N_{\max} + \pi/N_{\max}$ via independent derivation path)
- FRAMEWORK level, NOT theorem-grade per Calibration #27 STANDING
- Pending Cal cold-read for tier disposition

9. Coordination

Cal: REQUEST cold-read on Mechanism A FRAMEWORK with Calibration #27 STANDING discipline applied. Specific question: does the per-substrate-tick com-

mitment scale = α -quantum assumption have honest substrate-mechanism support, or is it Mode 1?

Keeper: K-202 K-audit pre-stage at FRAMEWORK tier per Cal #66 (NOT RATIFIED, NOT RIGOROUSLY CLOSED).

Grace: catalog Mechanism A FRAMEWORK entries at appropriate tier (FRAMEWORK).

Elie: per Keeper “build the next toy to verify a DERIVATION, not to verify a prediction-target” — Joos-Zeh γ framework toy ready for v0.4 Mechanism A hand-off when v0.5 substrate-tick scale derivation closes.

— Lyra, Task #320 v0.4 Mechanism A Deepening filed Sunday 2026-05-24 ~13:25 EDT per Casey “work the board” directive + Calibration #27 STANDING discipline applied.